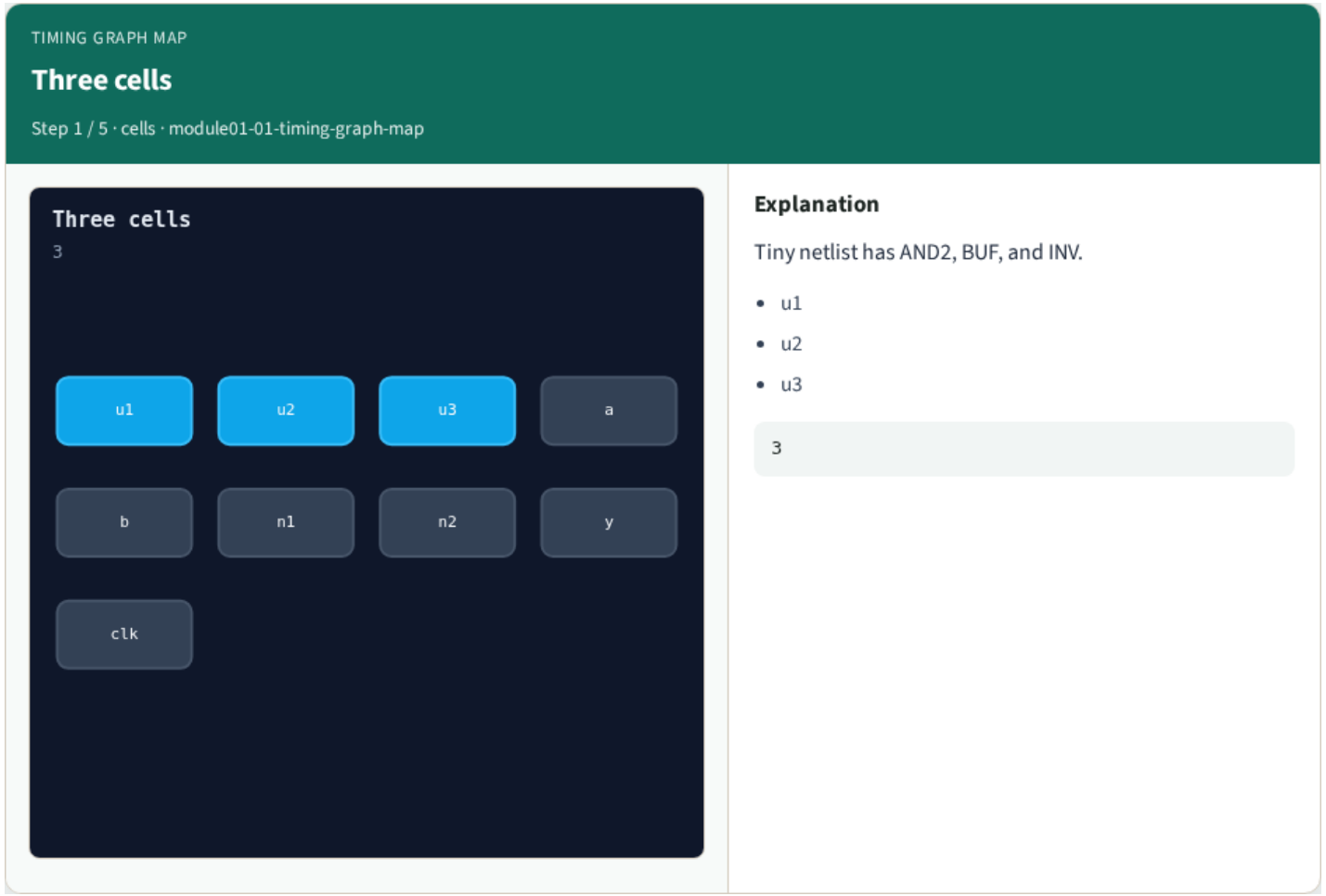
Why Timing

Map netlist cells to the OpenSTA timing graph

The idea

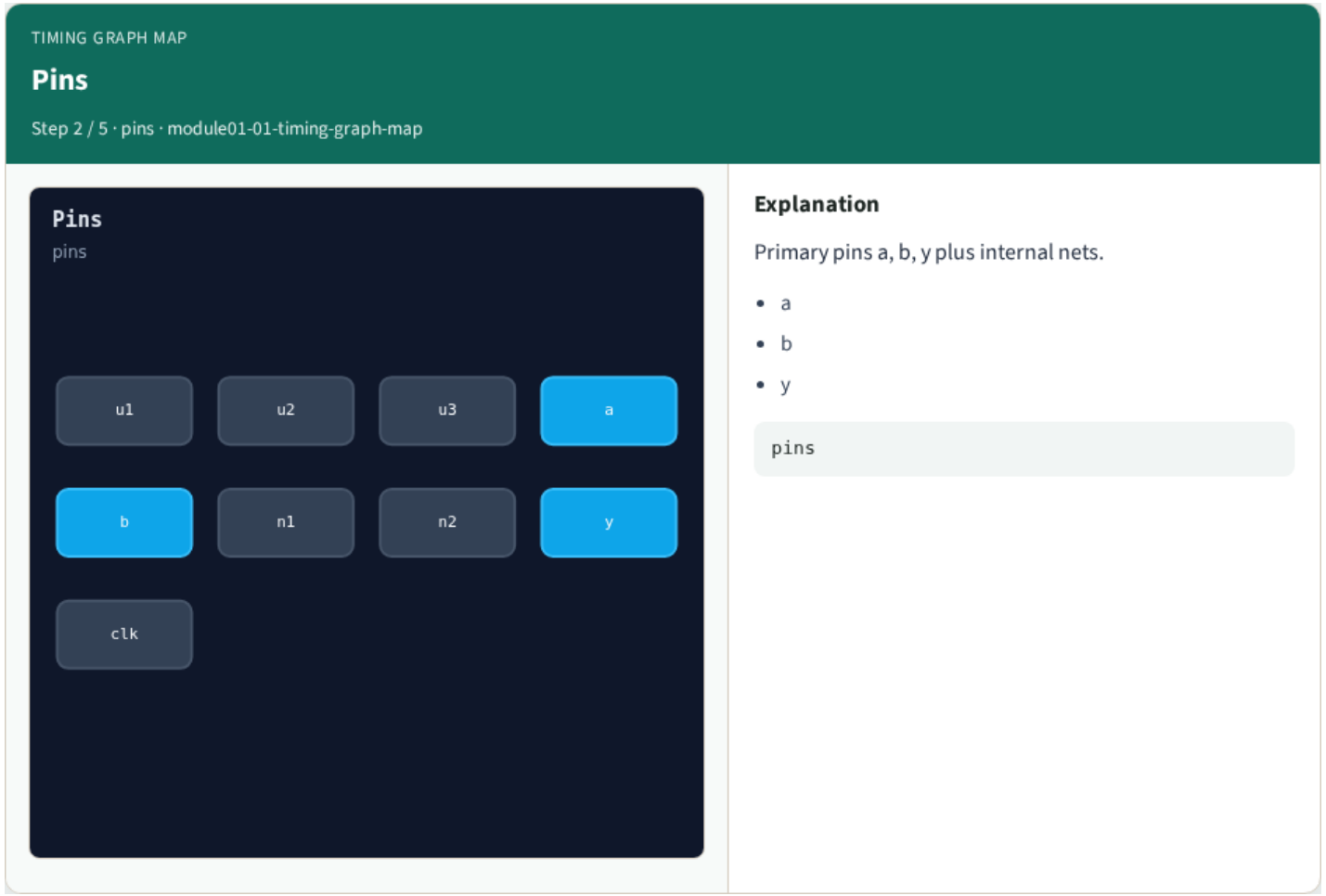
- GOLDEN: 3 cells u1/u2/u3; graph nodes = cells + pins

Three cells



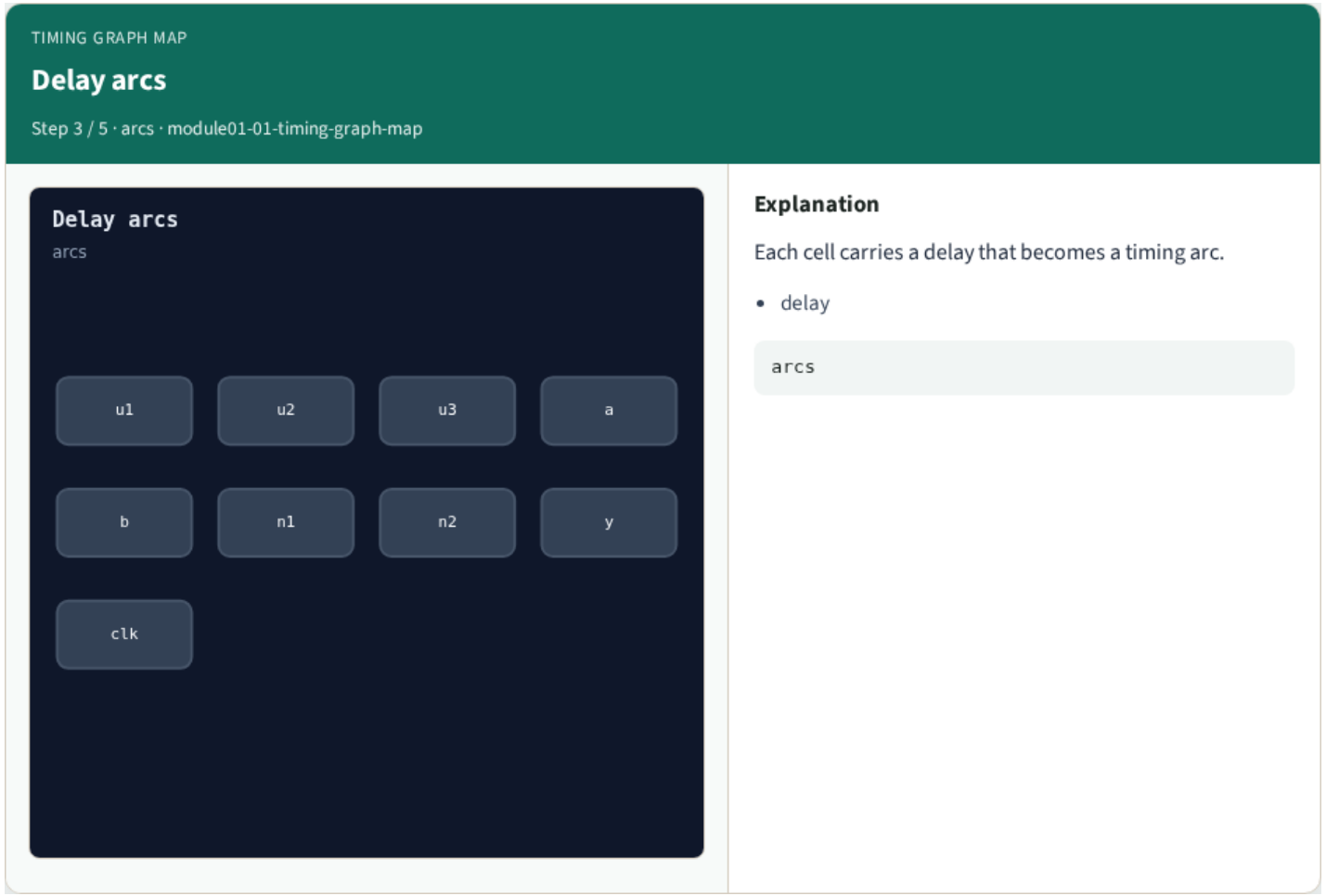
Three cells

Pins


TIMING GRAPH MAP
Pins
Step 2 / 5 · pins · module01-01-timing-graph-map
Pins
pins
u1
u2
u3
a
b
n1
n2
y
clk
Explanation
Primary pins a, b, y plus internal nets.
a
b
y
pins

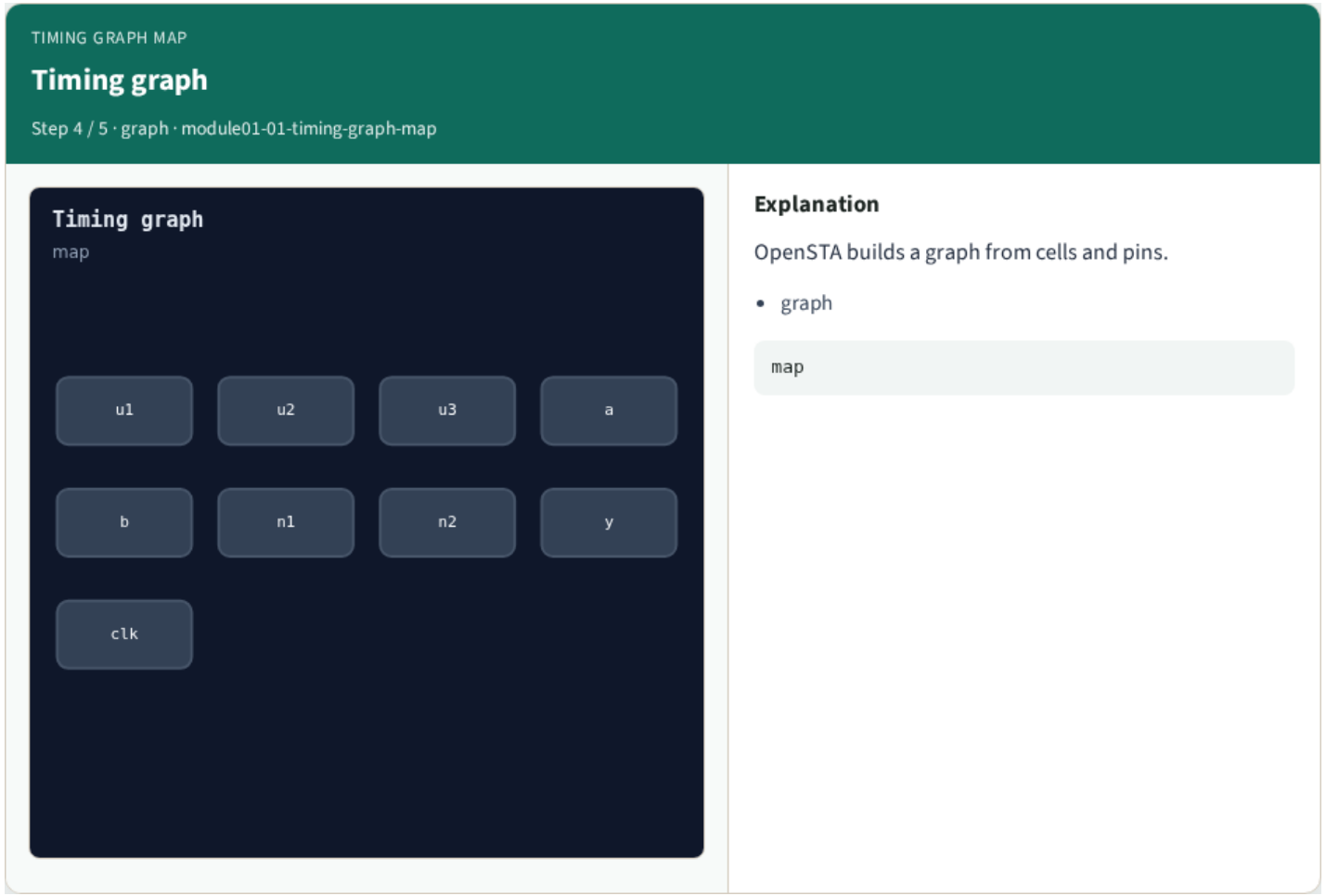
Pins

Delay arcs



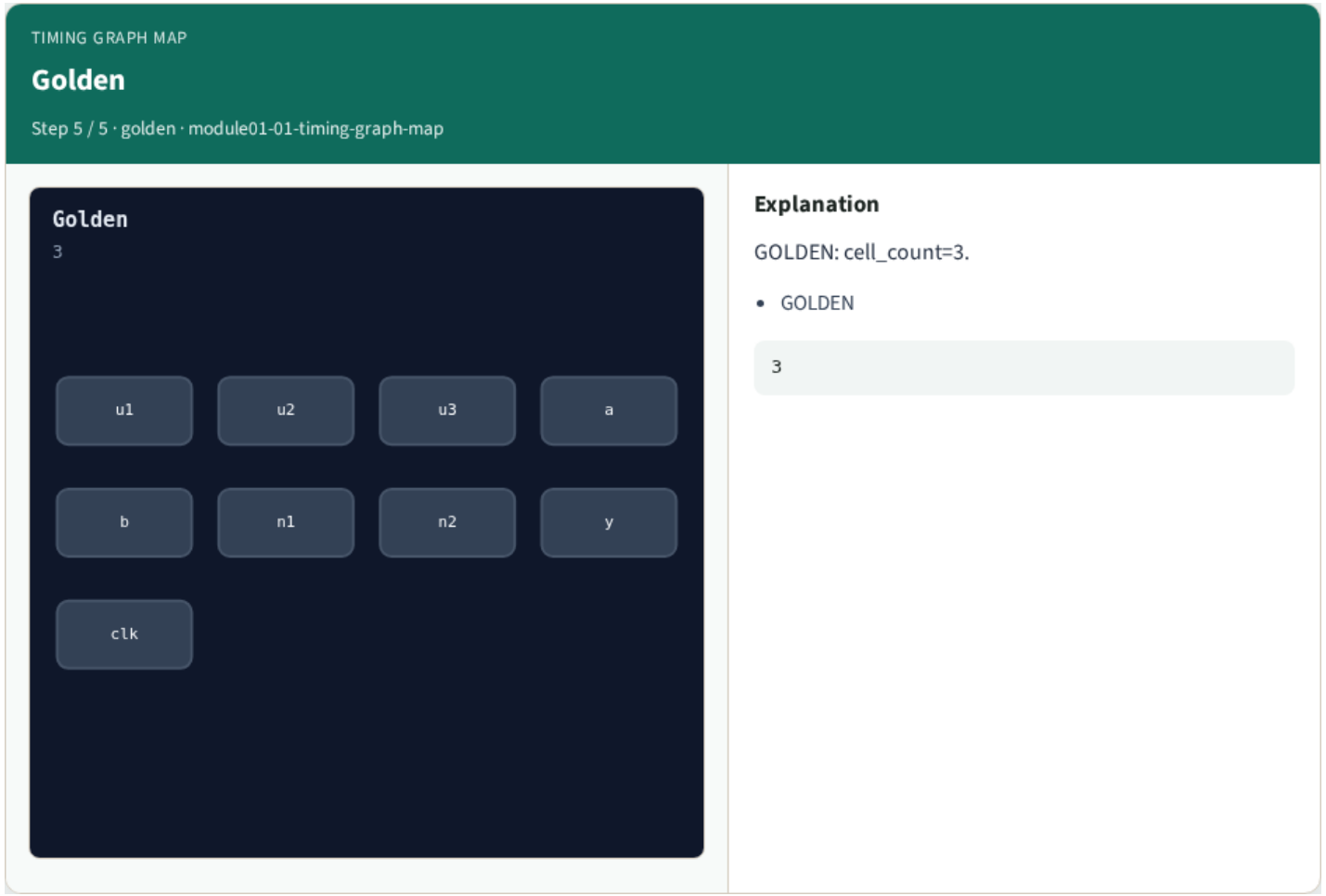
Delay arcs

Timing graph



Timing graph

Golden



Golden

Browser lab track

- Open **timing-graph-map**
- Run the stage buttons and Check the goldens

Implement track

- From common/, run solvers.py and confirm OPENSTA_MAP_OK

Pitfalls

- Mixing algorithm STA from learn_sta with OpenSTA command names
- Forgetting the clock period when reading slack

Try these

- From the common folder, run the OpenSTA map self-test

Try these — try these

```
cd courses/learn_opensta/common
python3 solvers.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
3 cells u1/u2/u3; graph nodes = cells + pins.
ok if goldens match
```

Next

- Complete the quiz for this part
- Continue the OpenSTA Part A map, then the real CLI track